



Binary Palindrome

In binary representation, a palindrome is a sequence of binary digits that reads the same forwards and backwards. It means that if you reverse the sequence, it will remain unchanged.

For $N = 16$, the binary representation is 10000. As you correctly pointed out, this binary sequence is not a palindrome because if we reverse it, we get 00001, which is not the same as the original sequence.

For $N = 17$, the binary representation is 10001. This binary sequence is a palindrome because if we reverse it, we still get 10001, which is the same as the original sequence.

In summary, the concept of binary palindromes involves examining whether a binary sequence remains unchanged when its digits are reversed. If it does, it is considered a binary palindrome.

1. Binary palindrome of N=5

- Binary: 101
- Reversed binary: 101
- Explanation: The binary sequence reads the same forwards and backwards.

2. Binary palindrome of N=51

- Binary: 110011
- Reversed binary: 110011
- Explanation: The binary sequence remains unchanged when reversed.

3. Binary palindrome of N=231

- Binary: 11100111
- Reversed binary: 11100111
- Explanation: The binary sequence is the same when reversed.

```
1. import java.util.*;
2. public class BinaryPalindromeChecker {
3.     public static boolean isBinaryPalindrome(int x) {
4.         int reversed = 0;
5.         int original = x;
6.         while (x > 0) {
7.             reversed <<= 1;           // Left shift by 1 bit
8.             reversed |= (x & 1);      // Add the least significant bit of 'x'
9.             x >>= 1;                 // Right shift by 1 bit
10.        }
11.        return reversed == original;
12.    }
13.    public static void main(String[] args) {
14.        Scanner sc = new Scanner(System.in);
15.        int x = sc.nextInt();
16.        if (isBinaryPalindrome(x)) {
17.            System.out.println(x + " has a binary palindrome representation.");
18.        } else {
19.            System.out.println(x + " does not have a binary palindrome
20.            representation.");
21.        }
22.    }
```

1. The `isBinaryPalindrome` function takes an integer `x` as input and returns a boolean indicating whether its binary representation is a palindrome.
2. The function initializes two variables:
 - `reversed` (initialized as 0) will store the reversed binary representation of `x`.
 - `original` will store the original value of `x` for comparison later.
3. The function enters a while loop that continues until `x` becomes 0:
 - In each iteration:
 - The `reversed` value is left-shifted by 1 bit using the `<<=` bitwise left shift operator.
 - The least significant bit of `x` (obtained using the `x & 1` bitwise AND operation) is added to `reversed` using the bitwise OR assignment operator `|=`. This operation sets the corresponding bit in `reversed` to the value of the least significant bit of `x`.
 - `x` is right-shifted by 1 bit using the `>>=` bitwise right shift operator.
4. Once the loop completes, the binary representation of `x` has been reversed and stored in the variable `reversed`.
5. The function checks if the reversed binary representation (`reversed`) is equal to the original value of `x` stored in the variable `original`. If they are equal, it means that the binary representation is a palindrome, so the function returns `true`. Otherwise, it returns `false`.
6. In the `main` method, you can test the function by passing different integers to `isBinaryPalindrome` and print the result accordingly.

The approach used in the program avoids converting the integer to a binary string and instead reverses the binary representation using bitwise operations. By doing so, the program achieves a more efficient solution with a time complexity of $O(\log N)$, where N is the value of the input integer.



THANK YOU